

Smart Warehouse Security & Environmental Monitoring System

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: ELECTRONICS AND COMMUNICATION ENGINEERING

of Bharathiar University



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION

PSG COLLEGE OF TECHNOLOGY

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COIMBATORE - 641 004

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Bona fide record of work done by

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Faculty guide

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Head of the Department

Certified that the candidate was examined in the viva-voce examination held on
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(Internal Examiner)

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(External Examiner)

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1. Introduction

Modern warehouses store a wide range of valuable goods such as electronics, pharmaceuticals, food materials and industrial components. Maintaining security and environmental safety inside such storage facilities is extremely important. Traditional warehouse monitoring systems rely heavily on manual supervision, which may lead to delays in identifying potential threats or environmental issues.

The **Smart Warehouse Security and Environmental Monitoring System** is designed to provide an automated and intelligent monitoring solution using an embedded microcontroller platform. The system continuously observes important conditions such as worker check-in status and environmental temperature levels. If any abnormal situation occurs, the system generates immediate alerts through visual indicators and wireless communication.

Hardware	Component
MCU Board	BGK-RA6E2 (Renesas RA6E2, Cortex-M33, 200MHz)
Temperature Sensor	HS4001 (I2C, PMOD2 Connector)
BLE Module	DA14531 (UART, PMOD1 Connector)
IDE & Framework	e ² Studio with FSP v6.4.0
Toolchain	GNU ARM Embedded 13.2.1

Fig 1.1 Hardware components

The proposed system is implemented using the **Renesas RA6E2 microcontroller**, which is based on the high-performance ARM Cortex-M33 architecture. Sensors and communication modules are integrated with the

microcontroller to collect environmental data and transmit notifications. The system utilizes a temperature sensor to monitor environmental conditions, an RTC module for time-based worker verification, and a Bluetooth Low Energy module for remote notifications.

By combining sensing, processing, and wireless communication technologies, the system provides an efficient real-time monitoring solution for warehouse environments. This project demonstrates how embedded systems can be used to enhance safety, reliability, and automation in industrial storage facilities.

2. Objective

The main objective of this project is to design and implement an embedded monitoring system capable of improving warehouse safety and operational accountability. The system aims to detect abnormal environmental conditions and ensure that workers perform scheduled inspection activities.

The specific objectives of this project include:

- To develop an embedded monitoring system using the Renesas RA6E2 microcontroller.
- To implement temperature monitoring using a digital sensor for environmental safety.
- To verify worker presence using a time-based check-in mechanism.
- To generate visual alerts using onboard LEDs when abnormal conditions are detected.
- To transmit alert messages wirelessly to the warehouse owner using Bluetooth communication.
- To demonstrate integration of sensors, microcontroller peripherals, and communication modules using Renesas development tools.

The project also aims to demonstrate the practical application of **embedded system design concepts**, including hardware interfacing, peripheral configuration, and real-time monitoring.

HARDWARE MODULES

3. Block Diagram and its Description

The block diagram represents the overall architecture of the Smart Warehouse Monitoring System. The central component of the system is the **RA6E2 microcontroller**, which coordinates all sensing, processing, and communication tasks.

The **temperature and humidity sensor (HS4001)** is connected to the microcontroller through the I2C communication interface. This sensor continuously measures environmental conditions inside the warehouse and sends digital data to the controller.

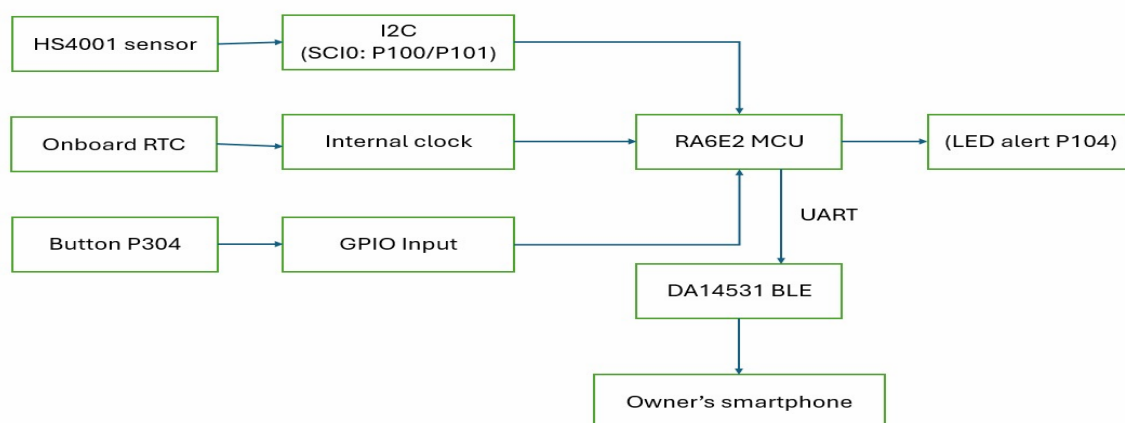


Fig 3.1 Block diagram

The **Real Time Clock (RTC)** module inside the microcontroller is used to track time and schedule worker check-in deadlines. If the worker fails to confirm presence within the specified time, the system triggers an alert.

A **push button** is used as a worker check-in interface. When pressed, it indicates that the worker has completed the inspection process.

The **BLE module (DA14531)** is connected through a UART interface and is responsible for transmitting alert messages to a smartphone or monitoring device.

The **LED indicator** provides visual alerts whenever the system detects abnormal conditions such as temperature threshold violations or missed worker check-ins.

Overall, the block diagram illustrates how different hardware modules interact with the microcontroller to form a complete monitoring system.

4. Circuit Diagram and its Description

The circuit diagram illustrates the electrical connections between the RA6E2 microcontroller and the external modules used in the system.

The **HS4001 sensor** is connected to the microcontroller through the I2C communication lines consisting of the Serial Clock Line (SCL) and Serial Data Line (SDA). These lines allow digital communication between the sensor and the controller.

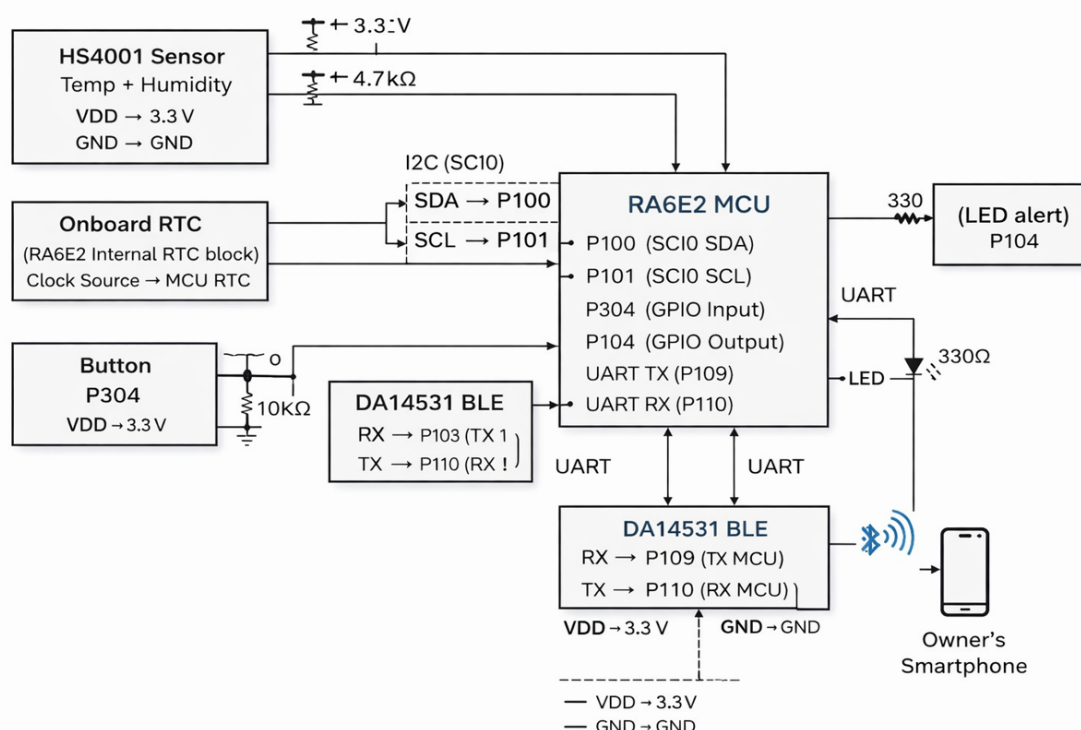


Fig 4.1 Circuit Diagram

The **BLE module** is connected using UART communication lines, which include the transmit (TX) and receive (RX) pins. Through this interface, the microcontroller sends alert messages that are broadcasted wirelessly.

A **push button switch** is connected to one of the GPIO input pins to allow workers to confirm their presence. When the button is pressed, the microcontroller receives an interrupt signal and records the check-in event.

An **LED indicator** is connected to an output GPIO pin. The microcontroller controls this LED to display warning signals whenever a threshold condition or alarm event occurs.

The circuit is powered through the development board's regulated power supply, ensuring stable operation of all connected modules.

5. Snapshots, Features and Description of Boards Used

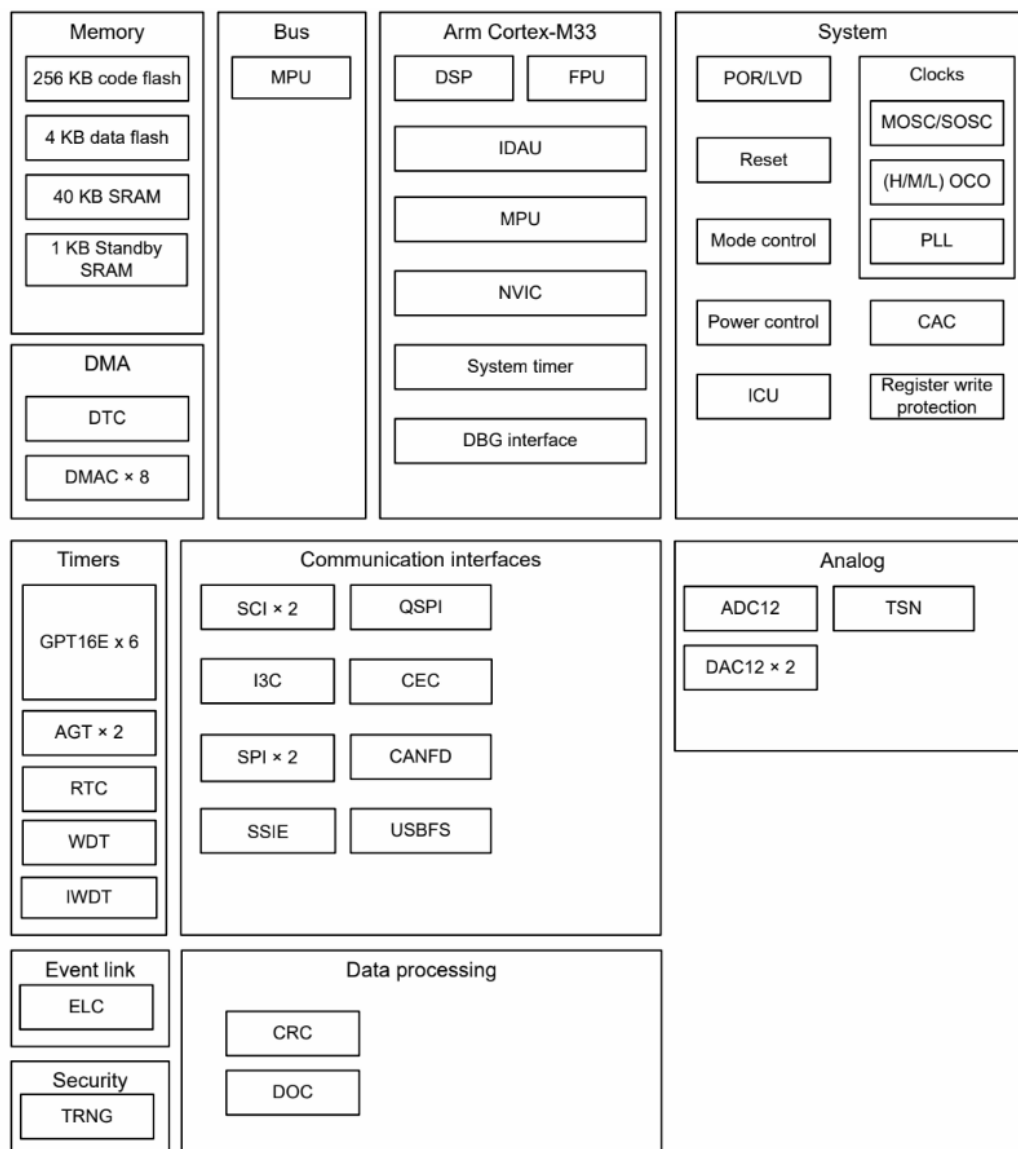


Fig 5.1 Block Diagram of RA6E2 Board

RA6E2 Microcontroller Board

The Renesas RA6E2 development board is the core hardware platform used for implementing the project. It is based on the ARM Cortex-M33 processor and is designed for secure embedded applications.

Key Features:

- ARM Cortex-M33 CPU running up to 200 MHz
- Built-in security features including TrustZone
- High performance with low power consumption
- Multiple communication interfaces such as UART, I2C, SPI
- Onboard debugging support
- Integrated clock and power management modules

The board provides expansion connectors that allow easy integration of sensors and communication modules.

HS4001 Temperature Sensor

The HS4001 sensor is used for monitoring environmental temperature and humidity levels within the warehouse.

Features:

- High accuracy temperature measurement
- Digital output through I2C interface
- Wide operating temperature range
- Low power consumption
- Factory calibration for reliable readings

The sensor provides real-time environmental data that is used to detect abnormal temperature conditions.



Fig 5.2 Product image of HS4001

BLE Module

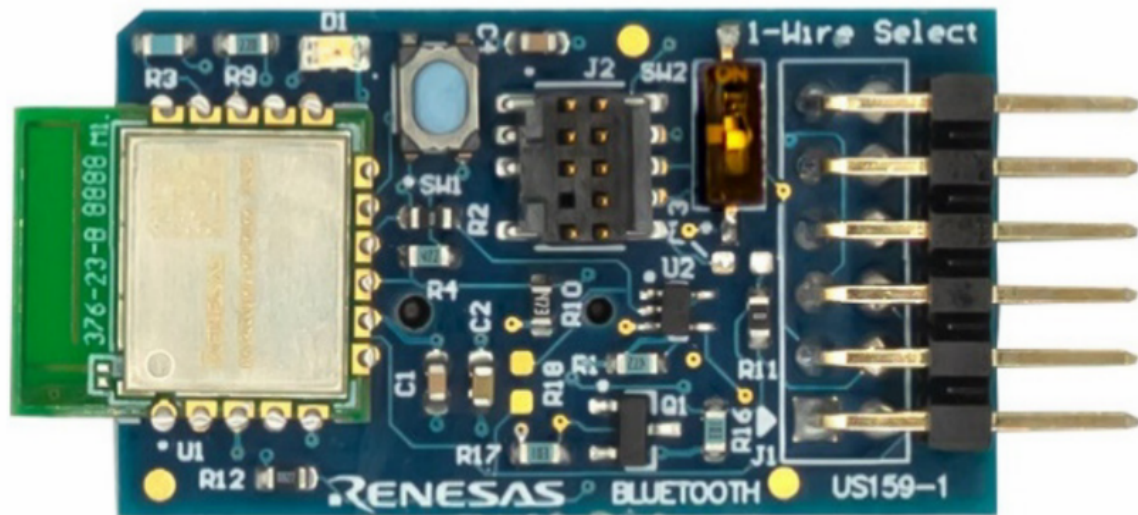


Fig 5.3 DA14531 Bluetooth Evaluation Module

BLE Module (DA14531)

The DA14531 module is a Bluetooth Low Energy communication device used for wireless data transmission.

Features:

- Supports Bluetooth 5.1 protocol
- Ultra-low power consumption
- UART communication interface
- Compact and efficient design
- Compatible with mobile applications for monitoring

The module enables remote monitoring by sending alerts directly to a smartphone.

SOFTWARE MODULES

6. Features of Quick Connect Studio

Quick Connect Studio is a graphical development environment provided by Renesas to simplify embedded system development. It allows developers to configure hardware peripherals and generate initialization code automatically.

Important features include:

- Visual configuration of microcontroller peripherals
- Automatic generation of driver code
- Integration with the Flexible Software Package (FSP)
- Simplified project creation and management
- Real-time debugging and monitoring support

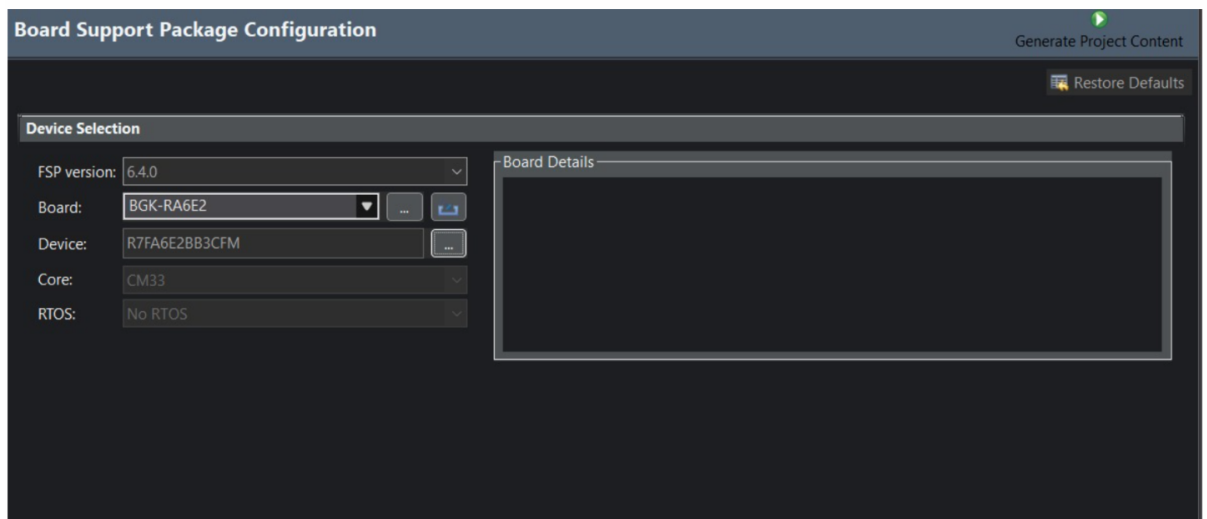
The tool significantly reduces development time by eliminating the need for manual configuration of hardware registers.

7. Project Development using Quick Connect Studio

The development of the Smart Warehouse Monitoring System was carried out using Quick Connect Studio along with the Renesas Flexible Software Package.

The development process includes:

1. Creating a new project using the RA6E2 board configuration.



2. Configuring required peripherals such as GPIO, UART, I2C, and RTC.

Interrupts Configuration

Generate Project Content

User Events

New User Event > Remove

EventISR

Allocations

InterruptEventISR

0RTC ALARM (Alarm interrupt)rtc_alarm_periodic_isr

1RTC CARRY (Carry interrupt)rtc_carry_isr

2SCI0 RXI (Receive data full)sci_uart_rx_isr

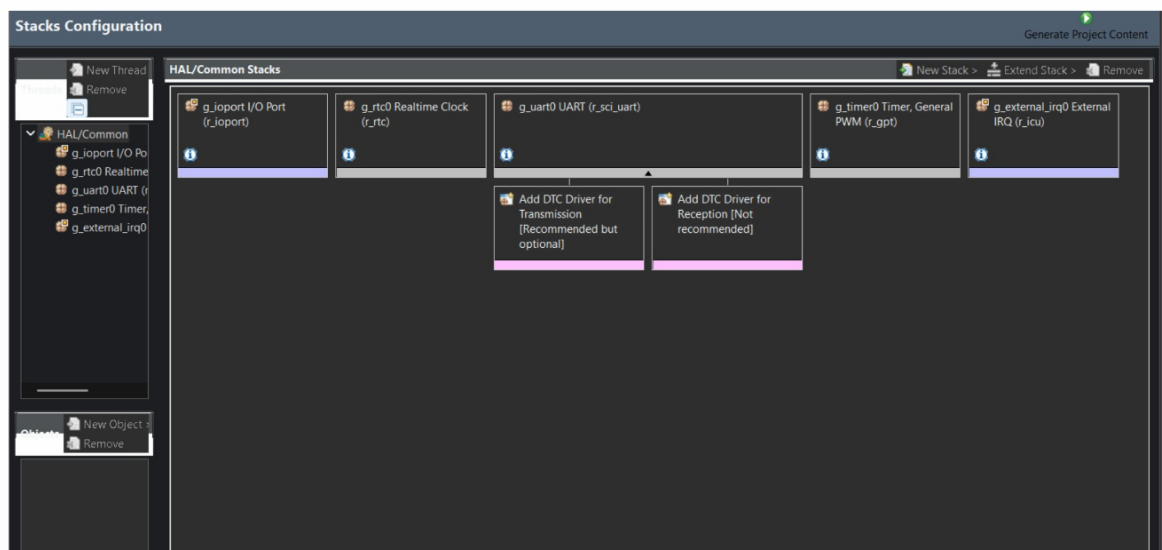
3SCI0 TXI (Transmit data empty)sci_uart_tx_isr

4SCI0 TEI (Transmit end)sci_uart_te_isr

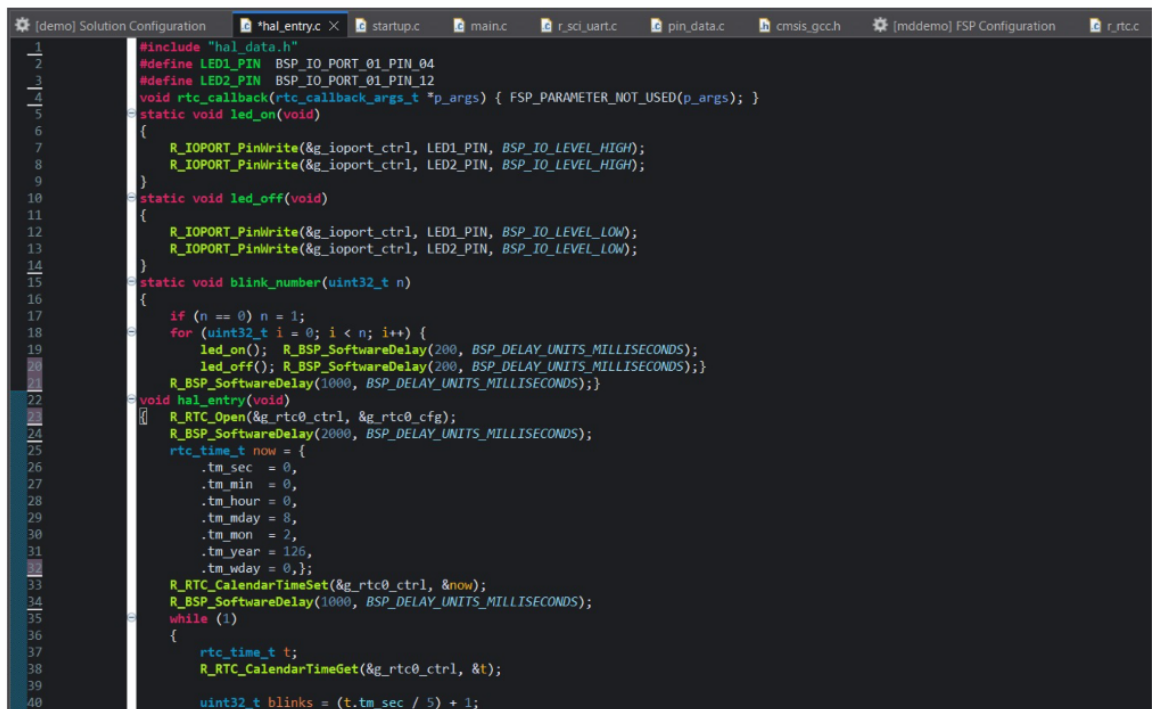
5SCI0 ERI (Receive error)sci_uart_eri_isr

6ICU IRQ0 (External pin interrupt 0)r_icu_isr

3. Adding sensor drivers and communication stacks through the FSP configuration tool.



4. Generating initialization code automatically.



```
1 #include "hal_data.h"
2 #define LED1_PIN BSP_IO_PORT_01_PIN_04
3 #define LED2_PIN BSP_IO_PORT_01_PIN_12
4 void rtc_callback(rtc_callback_args_t *p_args) { FSP_PARAMETER_NOT_USED(p_args); }
5 static void led_on(void)
6 {
7     R_IOPORT_PinWrite(&g_ioport_ctrl, LED1_PIN, BSP_IO_LEVEL_HIGH);
8     R_IOPORT_PinWrite(&g_ioport_ctrl, LED2_PIN, BSP_IO_LEVEL_HIGH);
9 }
10 static void led_off(void)
11 {
12     R_IOPORT_PinWrite(&g_ioport_ctrl, LED1_PIN, BSP_IO_LEVEL_LOW);
13     R_IOPORT_PinWrite(&g_ioport_ctrl, LED2_PIN, BSP_IO_LEVEL_LOW);
14 }
15 static void blink_number(uint32_t n)
16 {
17     if (n == 0) n = 1;
18     for (uint32_t i = 0; i < n; i++) {
19         led_on(); R_BSP_SoftwareDelay(200, BSP_DELAY_UNITS_MILLISECONDS);
20         led_off(); R_BSP_SoftwareDelay(200, BSP_DELAY_UNITS_MILLISECONDS);
21     }
22     R_BSP_SoftwareDelay(1000, BSP_DELAY_UNITS_MILLISECONDS);
23 }
24 void hal_entry(void)
25 {
26     R_RTC_Open(&g_rtc0_ctrl, &g_rtc0_cfg);
27     R_BSP_SoftwareDelay(2000, BSP_DELAY_UNITS_MILLISECONDS);
28     rtc_time_t now = {
29         .tm_sec = 0,
30         .tm_min = 0,
31         .tm_hour = 0,
32         .tm_mday = 8,
33         .tm_mon = 2,
34         .tm_year = 126,
35         .tm_wday = 0;
36     };
37     R_RTC_CalendarTimeSet(&g_rtc0_ctrl, &now);
38     R_BSP_SoftwareDelay(1000, BSP_DELAY_UNITS_MILLISECONDS);
39     while (1)
40     {
41         rtc_time_t t;
42         R_RTC_CalendarTimeGet(&g_rtc0_ctrl, &t);
43         uint32_t blinks = (t.tm_sec / 5) + 1;
44     }
```

5. Writing application logic to read sensor data and monitor system conditions.
6. Compiling and building the project using the embedded toolchain.
7. Flashing the compiled program into the microcontroller.

Summary	
Board:	BGK-RA6E2
Device:	R7FA6E2BB3CFM
Core:	CM33
Toolchain:	GCC ARM Embedded
Toolchain Version:	13.2.1.arm-13-7
FSP Version:	6.4.0
Project Type:	Flat
Location:	C:/Users/natar/e2_studio/workspace/mddemo ➡
Selected software components	
Arm CMSIS Version 6 - Core (M)	v6.1.0+fsp.6.4.0
RA6E2-BGK Board Support Files	v6.4.0
Board support package for RA6E2	v6.4.0
Board support package for R7FA6E2BB3CFM	v6.4.0
Board support package for RA6E2 - Events	v6.4.0
Board support package for RA6E2 - FSP Data	v6.4.0
Board support package for RA6E2 - Linker	v6.4.0
Board Support Package Common Files	v6.4.0
General PWM Timer	v6.4.0
External Interrupt	v6.4.0
I/O Port	v6.4.0
Real Time Clock	v6.4.0
SCI UART	v6.4.0
Common source for all template projects.	v6.4.0

This development workflow ensures efficient integration between hardware and software components.

8. Flowchart / Steps

System Operation Steps:

1. Initialize microcontroller peripherals and communication interfaces.
2. Configure the RTC module for worker check-in monitoring.
3. Start reading environmental data from the temperature sensor.
4. Compare the measured temperature with the predefined threshold value.
5. If temperature exceeds the threshold, trigger an alert.
6. Check whether the worker check-in button has been pressed within the scheduled time.
7. If the worker fails to check in, generate an alert notification.
8. Send warning messages to the smartphone using BLE communication.
9. Continue monitoring in a continuous loop.

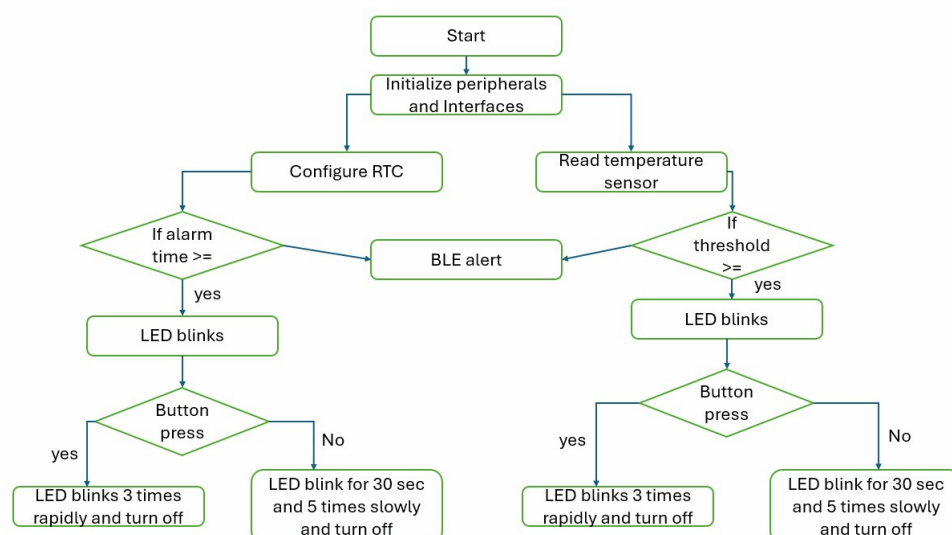


Fig 8.1 Flowchart

9. Library Files / API / BSP Files

Several software libraries and driver modules are used in the project to simplify hardware control.

Important software components include:

- **BSP (Board Support Package)** - Provides hardware initialization and board configuration settings.
- **GPIO Driver (r_ioport)** - Used for controlling LEDs and reading button inputs.
- **I2C Driver (r_sci_i2c)** - Enables communication with the HS4001 sensor.
- **UART Driver (r_sci_uart)** - Used for communication with the BLE module.
- **RTC Driver (r_rtc)** - Handles real-time clock operations for scheduling tasks.
- **Sensor Driver (rm_hs400x)** - Provides high-level functions for reading sensor data.

These APIs allow developers to interact with hardware peripherals through easy-to-use software functions.

10. Tools for Downloading .SREC to Microcontroller

The compiled program is converted into an **SREC (Motorola S-Record) file format**, which is commonly used for programming embedded microcontrollers.

The flashing process is performed using **SEGGER J-Flash Lite**, which is a reliable tool for programming Renesas devices.

Steps involved:

1. Connect the RA6E2 development board to the computer via USB.
2. Open the J-Flash Lite programming software.
3. Select the correct target microcontroller device.
4. Load the generated SREC file.

5. Program the device through the SWD interface.
6. Verify the flashing process to ensure successful programming.

11. Other Tools Used

Several development tools were used during the implementation of the project.

e² Studio IDE

This integrated development environment is used for writing, compiling, and debugging embedded programs.

Flexible Software Package (FSP)

FSP provides hardware abstraction layers and drivers that simplify peripheral configuration.

GNU ARM Toolchain

This toolchain compiles C code into machine code that can run on the ARM Cortex-M33 processor.

These tools together create a complete embedded development ecosystem.

HARDWARE SOFTWARE INTEGRATION

12. Working of the Project

When the system is powered on, the microcontroller initializes all required peripherals and sensors. The temperature sensor begins transmitting environmental data through the I2C interface.

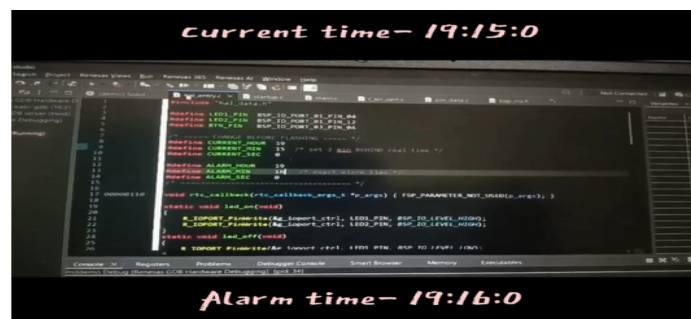


Fig 12.1 Hardware output

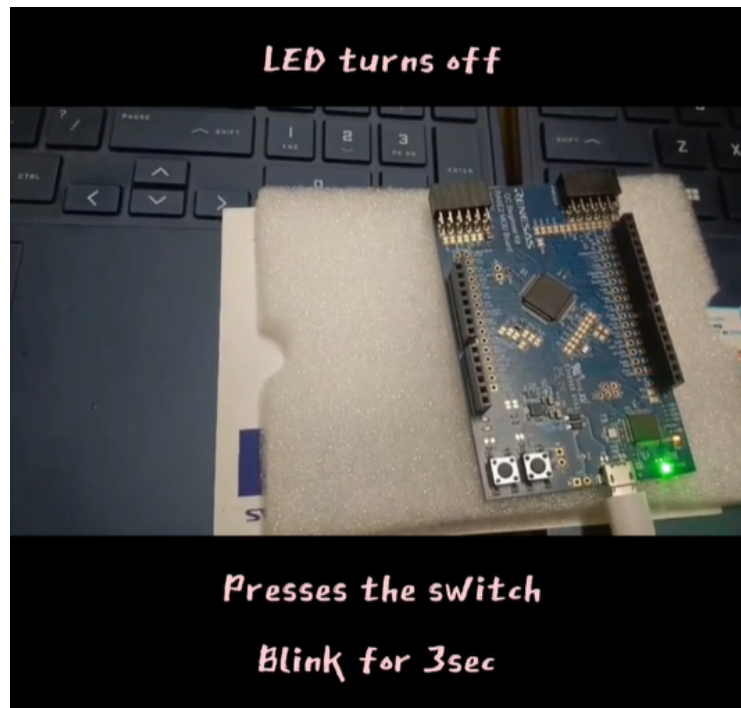


Fig 12.2 Case 1 of output



Fig 12.3 Case 2 of output

The microcontroller continuously monitors this data and compares it against predefined safety thresholds. If the temperature rises above the safe limit, the LED indicator begins blinking and a warning message is transmitted via the BLE module.

In the above example when 19.16.0 time is reached the led starts to blink for 30 sec if the worker presses the switch with in that 30 sec then the workers entry is marked ,if the worker fails to press the switch the led blinks for 30 sec and blinks again for extra 5 sec then turns off indicating that the worker is failed to enter with in the time.

Simultaneously, the RTC module tracks the worker inspection schedule. If the worker presses the check-in button before the deadline, the system records the event and resets the monitoring cycle. If the worker fails to check in within the specified time, an alert notification is generated.

The BLE module sends these alerts to a connected smartphone, allowing the warehouse owner to monitor the system remotely.

13. Use Cases to Justify Cortex-M33 and TrustZone

The Cortex-M33 processor is designed for secure and high-performance embedded applications. It provides hardware security features that are essential for modern IoT and industrial systems.

In this project, the Cortex-M33 architecture enables efficient sensor data processing and communication while maintaining low power consumption.

The TrustZone security feature allows the system to divide operations into secure and non-secure environments. Sensitive operations such as communication authentication and system control can be executed in the secure region, preventing unauthorized access.

This makes the platform suitable for future expansions involving secure cloud communication or remote device management.

14. Results

The Smart Warehouse Monitoring System was successfully implemented and tested using the RA6E2 microcontroller platform.

The system was able to:

- Accurately monitor environmental temperature conditions.
- Detect temperature threshold violations and trigger alerts.
- Record worker check-in events through button input.
- Send wireless notifications using Bluetooth communication.
- Provide visual warnings using LED indicators.

Testing confirmed that the system operates reliably under different environmental conditions.

15. Conclusion and Future Work

The project successfully demonstrates the implementation of a real-time warehouse monitoring system using an embedded microcontroller platform. The system integrates environmental sensing, worker verification, and wireless communication to provide a comprehensive safety solution.

The use of the Renesas RA6E2 microcontroller and its supporting development tools enabled efficient hardware and software integration. The system provides timely alerts and remote monitoring capabilities, improving warehouse management and operational safety.

Future improvements could include integration of additional sensors such as smoke detectors, motion sensors, and gas sensors. Cloud connectivity can also be implemented to enable remote monitoring through web dashboards and data analytics.

16. Appendix/ BIBLIOGRAPHY

The appendix section contains additional technical documents used during the development of the project.

Examples include:

- RA6E2 Microcontroller Datasheet
- HS4001 Sensor Datasheet
- DA14531 BLE Module Datasheet
- Development Board Schematics
- Software API Documentation

These references provide detailed technical specifications for the hardware and software components used in the project.

Contributions:

Hardware Implementation : Haripriya(23l116), Nataraj(23l234) ,
Nivedhita (23l138)

Bluetooth module : Tharshanaa(23l160)

Ppt : Haripriya(23l116), Bhadmavaisnavi (23l104),
Gopika(23l114)

Video : Haripriya(23l116),Bhadmavaisnavi(23l104)
Gopika(23l114)

Document : Viknesh(23l163), Lijith(23l235)

